PYTHON PROGRAMMING LANGUAGE

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Python Programming Language

STAGE 0 — Programming & Python Foundations

Goal

Understand what Python is and how it executes before writing code.

Topics

- What Programming Is
- What Python Is (Language vs Implementation)
- CPython Overview
- Interpreter vs Compiler
- Python Execution Model
- Python Source Files
- Python Versions and Release Cycle
- Installing Python Correctly
- Python REPL
- Running Scripts
- Code Editors and IDEs
- Syntax Rules
- Indentation and Code Blocks
- Comments and Documentation Strings

STAGE 1 — Core Syntax & Expressions

Goal

Write correct Python statements and expressions.

Topics

- Variables and Assignment
- Identifiers and Keywords
- Built-in Data Types Overview
- Dynamic Typing
- Type Checking
- Type Conversion
- Input Handling
- Output Formatting
- Operators
    - Arithmetic
    - Comparison
    - Logical
    - Assignment
    - Bitwise
    - Membership
    - Identity
- Expressions
- Operator Precedence

STAGE 2 — Python Object Model & Primitive Types

Goal

Understand Python's object-based nature.

Topics

- Everything Is an Object

- Object Identity
- Object Mutability
- Integers
- Floating-Point Numbers
- Complex Numbers
- Booleans
- NoneType
- Numeric Precision
- Immutability Concepts

STAGE 3 — Strings & Text Processing

Goal

Master text handling, a core Python skill.

Topics

- String Creation
- Indexing
- Slicing
- String Immutability
- Common String Methods
- Searching and Replacing
- String Formatting
    - f-strings
    - format()
- Escape Characters
- Unicode

- Encoding and Decoding
- Text Normalization

STAGE 4 — Control Flow & Logic

Goal

Control program execution accurately.

Topics

- Boolean Expressions
- Truthy and Falsy Values
- if Statements
- elif Chains
- else Blocks
- Nested Conditions
- match / case (Structural Pattern Matching)

STAGE 5 — Loops & Iteration

Goal

Repeat logic safely and efficiently.

Topics

- for Loops
- while Loops
- Loop Control Statements
    - break
    - continue
    - pass

- Nested Loops
- Loop else Clause
- Common Loop Patterns
- Avoiding Infinite Loops

STAGE 6 — Core Data Structures

Goal

Store, access, and manipulate data correctly.

Topics

- Lists
    - Creation
    - Indexing
    - Slicing
    - Methods
    - Mutability
- Tuples
    - Packing
    - Unpacking
    - Immutability
- Sets
    - Uniqueness
    - Mathematical Operations
- Dictionaries
    - Keys and Values
    - Hashing Basics

    - Dictionary Methods
- Choosing Data Structures

STAGE 7 — Functions & Scope

Goal

Write reusable and maintainable code.

Topics

- Function Definitions
- Calling Functions
- Parameters vs Arguments
- Return Values
- Default Arguments
- Keyword Arguments
- Variable-Length Arguments
- Scope Rules
- Local vs Global Scope
- LEGB Rule
- Docstrings
- Clean Function Design

STAGE 8 — Advanced Functions & Functional Concepts

Goal

Use Python’s functional capabilities correctly.

Topics

- Functions as First-Class Objects

- Lambda Functions
- map()
- filter()
- reduce()
- Recursion
- Closures
- Decorators
- Function Annotations

STAGE 9 — Error Handling & Robustness

Goal

Build fault-tolerant programs.

Topics

- Syntax Errors vs Runtime Errors
- Exceptions
- Built-in Exception Hierarchy
- try / except
- Multiple except Blocks
- else Clause
- finally Clause
- Raising Exceptions
- Custom Exceptions
- Error Design Best Practices

STAGE 10 — Modules, Packages & Environments

Goal

Structure real Python projects professionally.

Topics

- Python Modules
- import Mechanics
- from ... import
- Import Aliases
- Python Packages
- **init**.py
- Module Search Path
- Virtual Environments
    - venv
- Package Installation
    - pip
- pyproject.toml
- Dependency Management
- poetry
- uv
- pipenv (Legacy Awareness)

STAGE 11 — File Handling & OS Interaction

Goal

Work with real files and system resources.

Topics

- File Paths

- Pathlib
- File Modes
- Reading Files
- Writing Files
- Appending Files
- CSV Files
- JSON Files
- os Module
- sys Module
- Environment Variables

STAGE 12 — Object-Oriented Programming

Goal

Design structured and scalable programs.

Topics

- OOP Principles
- Classes and Objects
- Attributes
- Methods
- Constructors
- Instance Variables
- Class Variables
- Encapsulation
- Access Control
- Inheritance

- Method Overriding
- Polymorphism
- Abstraction
- Abstract Base Classes
- Magic Methods
- Dataclasses

STAGE 13 — Advanced OOP & Design

Goal

Apply professional design practices.

Topics

- Multiple Inheritance
- Method Resolution Order (MRO)
- Composition vs Inheritance
- Mixins
- SOLID Principles
- Design Patterns in Python
- Anti-Patterns

STAGE 14 — Python Standard Library

Goal

Use built-in tools instead of reinventing the wheel.

Topics

- math
- random

- datetime
- collections
- itertools
- functools
- statistics
- enum
- typing
- copy
- pickle
- argparse
- logging

STAGE 15 — Iterators, Generators & Context Managers

Goal

Write memory-efficient and elegant code.

Topics

- Iterable vs Iterator
- Iterator Protocol
- Generators
- yield
- Generator Expressions
- Context Managers
- with Statement
- Custom Context Managers

STAGE 16 — Memory, Performance & Internals

Goal

Understand Python at runtime level.

Topics

- Python Memory Model
- Stack vs Heap
- Reference Counting
- Garbage Collection
- Shallow vs Deep Copy
- Mutability Pitfalls
- Time Complexity
- Space Complexity
- Profiling
- Optimization Techniques

STAGE 17 — Concurrency & Async Programming

Goal

Handle multiple tasks efficiently.

Topics

- Processes vs Threads
- Global Interpreter Lock (GIL)
- threading
- multiprocessing
- concurrent.futures
- Async Programming Model

- asyncio
- Event Loop
- Async Functions
- Async Context Managers

STAGE 18 — Testing, Quality & Tooling

Goal

Ensure correctness and maintainability.

Topics

- Testing Principles
- Unit Testing
- pytest
- Assertions
- Fixtures
- Mocking
- Code Coverage
- Linting
- Formatting
- PEP 8
- Type Hinting
- Static Type Checking (mypy / pyright)
- Ruff / Black

STAGE 19 — Packaging, Distribution & Maintenance

Goal

Ship Python software properly.

Topics

- Project Layout Standards
- Packaging Workflows
- Versioning
- Dependency Locking
- Wheels
- Publishing Packages
- Backward Compatibility
- Deprecation Strategy

STAGE 20 — Senior Python Engineering

Goal

Operate as a senior Python engineer.

Topics

- Writing Production-Grade Python
- Reading CPython Source (High Level)
- Performance Trade-offs
- Refactoring Strategies
- Debugging Complex Systems
- Code Reviews
- Documentation Standards
- Technical Decision Making
- Mentorship
- Long-Term Maintenance

uzb:

Python dasturlash tili

BOSQICH 0 — Dasturlash va Python asoslari

Maqsad

Python nima va qanday ishlashini tushunish.

Mavzular

- Dasturlash nima
- Python nima (til va interpreter farqi)
- CPython haqida qisqacha
- Interpreter va kompilator farqi
- Python dasturining ishlash jarayoni
- Python fayllari va strukturalari
- Python versiyalari va yangilanishlar
- Python o‘rnatish
- Python REPL bilan ishlash
- Skriptni ishga tushirish
- Kod muharrirlari va IDE
- Sintaksis qoidalari
- Indentatsiya va kod bloklari
- Izohlar va hujjat yozish

BOSQICH 1 — Asosiy sintaksis va ifodalar

Maqsad

To‘g‘ri Python kodini yozishni o‘rganish.

Mavzular

- O‘zgaruvchilar va qiymat tayinlash
- Identifikatorlar va kalit so‘zlar
- Ma’lumot turlari (overview)
- Dinamik tur
- Tur tekshirish
- Tur konvertatsiyasi
- Foydalanuvchi kiritishi va chiqishi
- Operatorlar:
    - Arifmetik
    - Solishtirish
    - Mantiqiy
    - Tayinslash
    - Bitwise
    - A’zolik (membership)
    - Identifikatorlar (identity)
- Ifodalar va operator ustuvorligi

BOSQICH 2 — Python obyekti va primitive turlar

Maqsad

Python obyekt modeli va asosiy turlarni tushunish.

Mavzular

- Hamma narsa obyekt
- Obyekt identifikatori
- Obyekt o‘zgarmasligi va o‘zgarmas turlar

- Integer, Float, Complex
- Boolean
- NoneType
- Numerik aniqlik

BOSQICH 3 — Matnlar bilan ishlash

Maqsad

Matn bilan ishlashni puxta o‘rganish.

Mavzular

- String yaratish
- Index va slicing
- String o‘zgarmasligi
- String metodlari
- Qidirish va almashtirish
- Formatlash:
  - f-string
  - format()
- Escape belgilar
- Unicode va kodlash
- Matn normalizatsiyasi

BOSQICH 4 — Shartli operatorlar va mantiq

Maqsad

Dastur oqimini nazorat qilish.

Mavzular

- Boolean ifodalar
- True/Falsy qiymatlar
- if, elif, else
- Nested shartlar
- match / case (pattern matching)

BOSQICH 5 — Sikllar va iteratsiya

Maqsad

Kod takrorlash va boshqarish.

Mavzular

- for sikl
- while sikl
- break, continue, pass
- Nested sikllar
- sikl else bloki
- Mashhur sikl naqshlari
- Cheksiz sikllardan saqlanish

BOSQICH 6 — Asosiy ma’lumot tuzilmalari

Maqsad

Ma’lumotni saqlash va ishlash.

Mavzular

- List
    - yaratish, indekslash, slicing
    - metodlar, o‘zgaruvchanligi

- Tuple
    - packing/unpacking
    - o‘zgarmaslik
- Set
    - noyob elementlar
    - matematik operatsiyalar
- Dictionary
    - Keys/values
    - Hashing
    - metodlar
- Ma’lumot tuzilmasini tanlash

BOSQICH 7 — Funksiyalar va scope

Maqsad

Qayta ishlatiladigan va toza kod yozish.

Mavzular

- Funksiya yaratish va chaqirish
- Parametrlar va argumentlar
- Return qiymatlar
- Default va keyword argumentlar
- Variable-length argumentlar
- Scope: Local/Global
- LEGB qoidasi
- Docstring
- Toza funksiya dizayni

BOSQICH 8 — Advanced functions & Functional concepts

Maqsad

Python funksional imkoniyatlarini puxta o‘rganish.

Mavzular

- Funksiyalar birinchi darajali obyekt
- Lambda funksiyalar
- map, filter, reduce
- Recursion
- Closures
- Decorators
- Function annotations

BOSQICH 9 — Exception va xatolarga chidamlilik

Maqsad

Xatolardan xavfsiz ishlash.

Mavzular

- Syntax vs Runtime errors
- Exception turlari
- try / except
- Multiple except
- else/finally
- Custom exceptions
- Raise

BOSQICH 10 — Modul, package va muhit

Maqsad

Katta loyihalar uchun strukturani tushunish.

Mavzular

- Modul va import
- from ... import
- Import alias
- Package va **init**.py
- Module path
- Virtual environment: venv
- Pip
- pyproject.toml
- Poetry, UV, pipenv (legacy)

BOSQICH 11 — File va OS bilan ishlash

Maqsad

Fayl va tizim bilan ishlash.

Mavzular

- File paths va pathlib
- File modes
- Read, write, append
- CSV va JSON
- os va sys
- Environment variables

BOSQICH 12 — OOP asoslari

Maqsad

Katta va murakkab dasturlar yozish.

Mavzular

- Classes & Objects
- Attributes, Methods
- Constructor
- Instance va class variables
- Encapsulation, Access control
- Inheritance, Overriding
- Polymorphism
- Abstraction & ABC
- Magic methods, Dataclasses

BOSQICH 13 — Advanced OOP & Design

Maqsad

Professional dizayn va arxitekturani qo‘llash.

Mavzular

- Multiple inheritance & MRO
- Composition vs Inheritance
- Mixins
- SOLID principles
- Design patterns
- Anti-patterns

BOSQICH 14 — Python Standard Library

Maqsad

Built-in vositalardan foydalanish.

Mavzular

- math, random, datetime
- collections, itertools, functools
- statistics, enum, typing
- copy, pickle
- argparse, logging

BOSQICH 15 — Iterators, generators & context managers

Maqsad

Xotira samaradorligi va elegant kod.

Mavzular

- Iterable vs iterator
- Iterator protocol
- Generators, yield
- Generator expressions
- Context manager & with
- Custom context managers

BOSQICH 16 — Memory va performance

Maqsad

Python ichki ishlashini tushunish.

Mavzular

- Memory model
- Stack vs Heap
- Reference counting
- Garbage collection
- Shallow vs deep copy
- Time & space complexity
- Profiling va optimization

BOSQICH 17 — Concurrency & Async

Maqsad

Ko‘p vazifali dasturlar yozish.

Mavzular

- Threads vs Processes
- GIL
- threading, multiprocessing
- concurrent.futures
- Asyncio, Event loop
- Async functions & context managers

BOSQICH 18 — Testing va code quality

Maqsad

Dastur to‘g‘ri va saqlash oson bo‘lishi.

Mavzular

- Unit testing
- pytest

- Assertions, Fixtures
- Mocking
- Code coverage
- Linting & formatting
- PEP8
- Type hints
- Static type check (mypy/pyright)
- Black/Ruff

BOSQICH 19 — Packaging & distribution

Maqsad

Python dasturini tarqatish va boshqarish.

Mavzular

- Project layout
- Packaging
- Versioning
- Dependency locking
- Wheel
- Publishing
- Backward compatibility
- Deprecation strategy

BOSQICH 20 — Senior Python engineer

Maqsad

Professional Python muhandisi bo‘lish.

Mavzular

- Production-grade Python
- CPython source high-level
- Performance trade-offs
- Refactoring strategies
- Debugging complex systems
- Code review
- Documentation
- Technical decisions
- Mentorship
- Large codebase maintenance

🔹 **Backend va AI mosligi**

- **Backend**: Async, OOP, packaging, testing, concurrency, logging → barcha Python core bor
- **AI**: Data structures, memory efficiency, generators, performance, functional programming → barcha Python core bor
- Frameworklar (FastAPI, Django, PyTorch, NumPy) → bu roadmap ustiga qo‘shiladi, ammo Python asoslari bo‘lmasa ishlamaydi.

O’quv davri davomiyligi:

Bosqichlar va vaqt taxmini

Bosqich	%	Taxminiy o‘rganish soati
0. Python asoslari	0–5%	4 soat (2 dars)
1. Sintaksis, operatorlar	5–10%	6 soat (3 dars)
2. Primitive turlar	10–15%	6 soat (3 dars)
3. Strings	15–20%	6 soat (3 dars)
4. If/else	20–25%	4 soat (2 dars)
5. Loops	25–30%	6 soat (3 dars)
6. Data structures	30–35%	8 soat (4 dars)
7. Functions	35–40%	8 soat (4 dars)
8. Advanced functions	40–45%	6 soat (3 dars)
9. Exception handling	45–50%	4 soat (2 dars)
10. Modules & env	50–55%	6 soat (3 dars)
11. File & OS	55–60%	6 soat (3 dars)
12. OOP asoslari	60–70%	12 soat (6 dars)
13. Advanced OOP	70–75%	6 soat (3 dars)
14. Std library	75–80%	6 soat (3 dars)
15. Iterators, generators	80–85%	6 soat (3 dars)
16. Memory & performance	85–87%	4 soat (2 dars)
17. Concurrency & Async	87–90%	8 soat (4 dars)

Bosqich	%	Taxminiy o‘rganish soati
18. Testing & code quality	90–95%	6 soat (3 dars)
19. Packaging & distribution	95–98%	4 soat (2 dars)
20. Senior engineer skills	98–100%	6 soat (3 dars)

Umumiy soat va darslar

- Umumiy vaqt: **120 soat**
- Har dars 2 soat, haftada 3 dars → **haftada 6 soat**
- 120 / 6 = **20 hafta**
- **20 hafta ≈ 5 oy**

va

- Agar haftada **3 kun, har dars 2 soat** bo‘lsa:
  Python ni **taxminan 5 oyda** tugatish mumkin.
- Bu hisobda amaliy mashqlar va kichik testlar kiritilgan.